



Incorporating AI into Teaching

Kerry Back
Rice University
FMA 2025



JPMorgan: Chatbots to Agents

CNBC, September 30, 2025: In 2023, JPMorgan gave employees access to OpenAI's models through LLM Suite.

It was essentially a corporate ChatGPT tool used to draft emails and summarize documents.

About 250,000 JPMorgan employees have access to the platform today ... Half of them use it roughly every day.

JPMorgan is now early in the next phase of its AI blueprint: It has begun deploying agentic AI to handle complex multistep tasks for employees, according to an internal road map provided by the bank.

From Makers to Checkers

Derek Waldron, JP Morgan Chief Analytics Officer:

What we're working towards is that every employee will have their own personalized AI assistant; every process is powered by AI agents, and every client experience has an AI concierge.

You'll still have people at the top who are managing and have relationships with clients, but many, many of the processes underneath are now being done by AI systems.

Workers would shift from being creators of reports or software updates, or 'makers' ... to 'checkers' or managers of AI agents doing that work.



MIT/Media Labs Report

Pilots to Production

Only 5% of AI pilots reach production.

Used by Everyone

Workers from **over 90%** of companies surveyed reported regular use of personal AI tools for work tasks.

Almost every single person used an LLM in some form for their work.

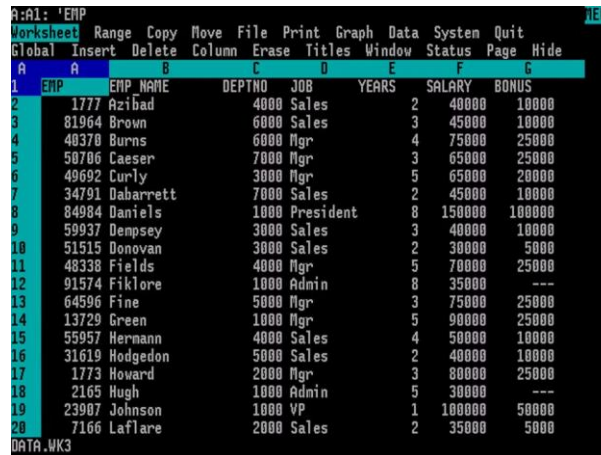
Change was only at the margins for the past 40 years



VisiCalc spreadsheet showing a list of items and their costs. The interface is green and black. The data is as follows:

ITEM	NO.	UNIT	COST
MUCK RAKE	43	12.95	556.85
BUZZ CUT	15	6.75	101.25
TOE TONER	250	49.95	12487.50
EYE SNUFF	2	4.95	9.90
SUBTOTAL			13155.50
9.75% TAX			1282.66
TOTAL			14438.16

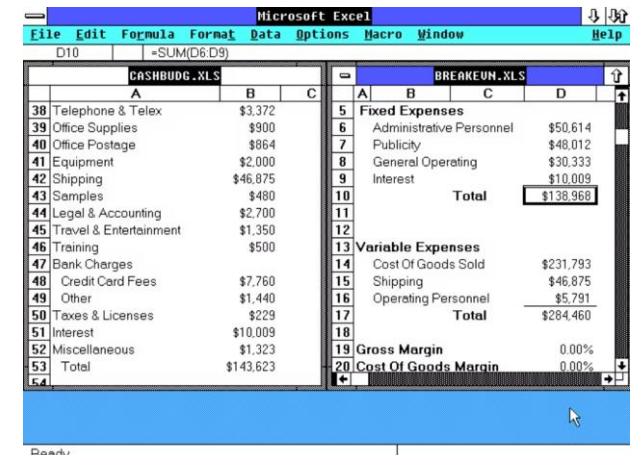
VisiCalc (1979)



Lotus 1-2-3 spreadsheet showing employee data. The interface is black and white. The data is as follows:

EMP	EMP NAME	DEPTNO	JOB	YEARS	SALARY	BONUS
1777	Azibad	4000	Sales	2	40000	10000
81964	Brown	6000	Sales	3	45000	10000
40370	Burns	6000	Mgr	4	75000	25000
50706	Caesar	7000	Mgr	3	65000	25000
49692	Curly	3000	Mgr	5	65000	20000
34791	Dabarrett	7000	Sales	2	45000	10000
84984	Daniels	1000	President	8	150000	100000
59937	Dempsey	3000	Sales	3	40000	10000
51515	Donovan	3000	Sales	2	30000	5000
48338	Fields	4000	Mgr	5	70000	25000
91574	Fiklore	1000	Admin	8	35000	---
64596	Fine	5000	Mgr	3	75000	25000
13729	Green	1000	Mgr	5	90000	25000
55957	Hermann	4000	Sales	4	50000	10000
31619	Hodgedon	5000	Sales	2	40000	10000
1773	Howard	2000	Mgr	3	80000	25000
2165	Hugh	1000	Admin	5	30000	---
23907	Johnson	1000	VP	1	100000	50000
7166	Laflare	2000	Sales	2	35000	5000

Lotus 1-2-3 (1983)



Microsoft Excel spreadsheet showing cash budget and breakeven analysis. The interface is blue and white. The data is as follows:

CASHBUDG.XLS			BREAKEUN.XLS		
A	B	C	A	B	C
38	Telephone & Telex	\$3,372	5	Fixed Expenses	
39	Office Supplies	\$900	6	Administrative Personnel	\$50,614
40	Office Postage	\$864	7	Publicity	\$48,012
41	Equipment	\$2,000	8	General Operating	\$30,333
42	Shipping	\$46,875	9	Interest	\$10,009
43	Samples	\$480	10	Total	\$138,968
44	Legal & Accounting	\$2,700	11		
45	Travel & Entertainment	\$1,350	12		
46	Training	\$500	13	Variable Expenses	
47	Bank Charges	---	14	Cost Of Goods Sold	\$231,793
48	Credit Card Fees	\$7,760	15	Shipping	\$46,875
49	Other	\$1,440	16	Operating Personnel	\$5,791
50	Taxes & Licenses	\$229	17	Total	\$284,460
51	Interest	\$10,009	18		
52	Miscellaneous	\$1,323	19	Gross Margin	0.00%
53	Total	\$143,623	20	Cost Of Goods Margin	0.00%
54			21		

Excel (1987)

For 40 years, the fundamental tool of business analysis remained essentially the same.



Now We're All Learning Together

AI has created an unprecedented situation:

Faculty, students, and industry professionals are all navigating the technological shift simultaneously.

Roadmap for Today's Discussion

A Course on AI for Finance

Chatbots and Agents

Previous Disruptive Technologies

AI in Finance Courses

AI in Education in General

Deep Dive into Claude Code





A Course on AI for Finance

Core Themes

Chatting:

- Treat AI as a colleague: collaborate, evaluate, iterate
- Coding: LLMs are unreliable for arithmetic; coding is essential

Building:

- Apps: to reuse tested code
- Custom chatbots: system prompts, RAG, or fine tuning
- Agents: chatbots with tools



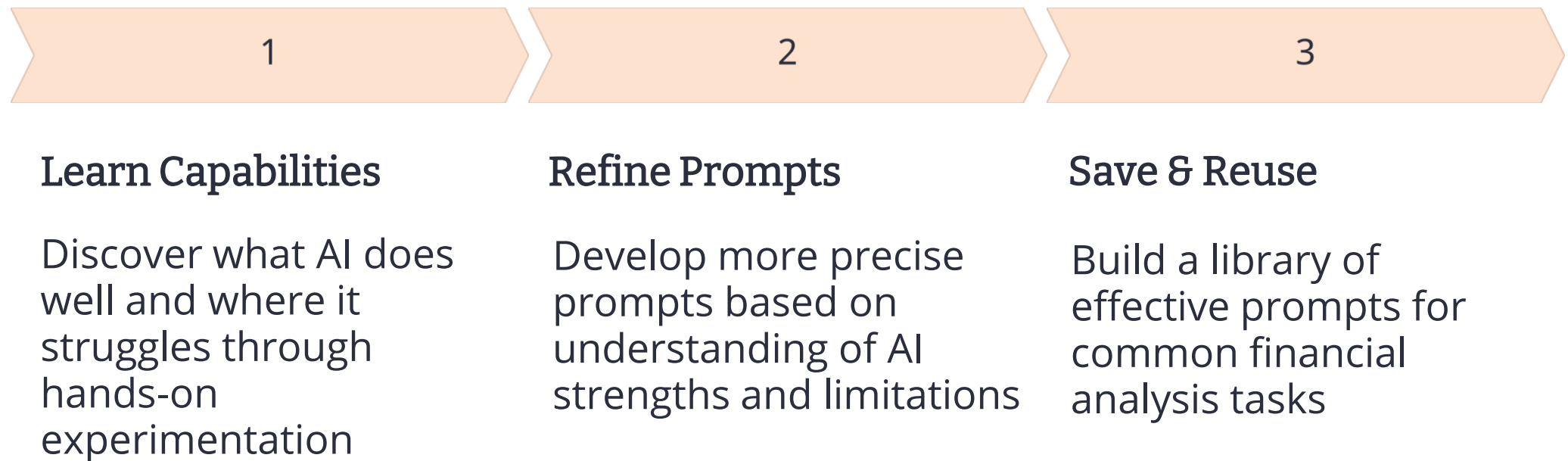
Prompt Engineering

Ethan Mollick, Wharton:

Treat AI like an infinitely patient new coworker ... As it is a coworker, you want to work with it, not just give it orders, and you also want to learn what it is good or bad at ... Working with AI is a dialogue, not an order.

Students ask how to prompt, but **practicing collaboration** is the key.

Learning How to Work with AI

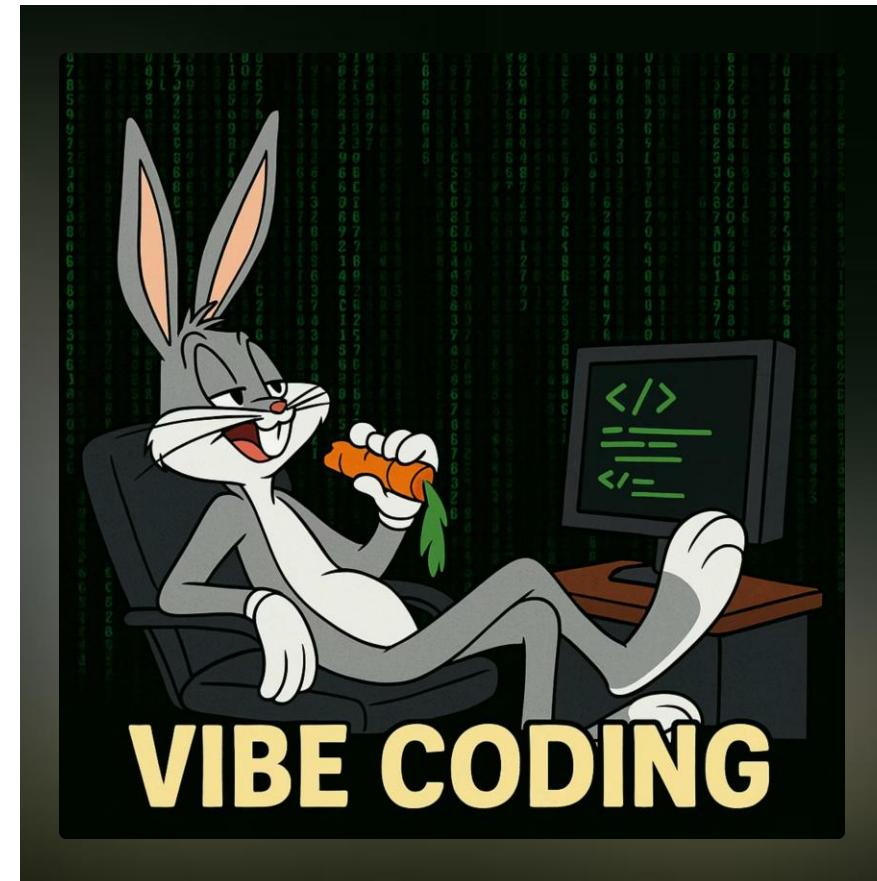


Class Exercise: Chat with a chatbot about financial analysis, evaluate the output, then develop a reusable prompt that achieves better results faster for similar future analyses.

Using AI to Write Code

Y Combinator CEO Garry Tan and Managing Partner Jared Friedman, March 2025:

For roughly a quarter of the startups in our Winter 2025 cohort, **95% of the codebase was written by AI.**





Claude + Python at Bridgewater

Aaron Linsky, CTO, AIA Labs at Bridgewater:

We've been developing capabilities powered by Claude since 2023 within AIA Labs.

Claude powered the first versions of our Investment Analyst Assistant, which streamlined our analysts' workflow by generating **Python code**, creating **data visualizations**, and iterating through **complex financial analysis tasks** with the precision of a junior analyst."

Coding Tools

Chatbots with Code Execution

ChatGPT, Gemini, Claude, **Julius.ai** - integrated environments for coding and testing

Python Environments with AI

Google Colab, VS Code, Cursor, Windsurf - traditional development enhanced by AI assistance

All-in-One Solutions

Replit - complete development and hosting platform with AI integration

My Current Preference

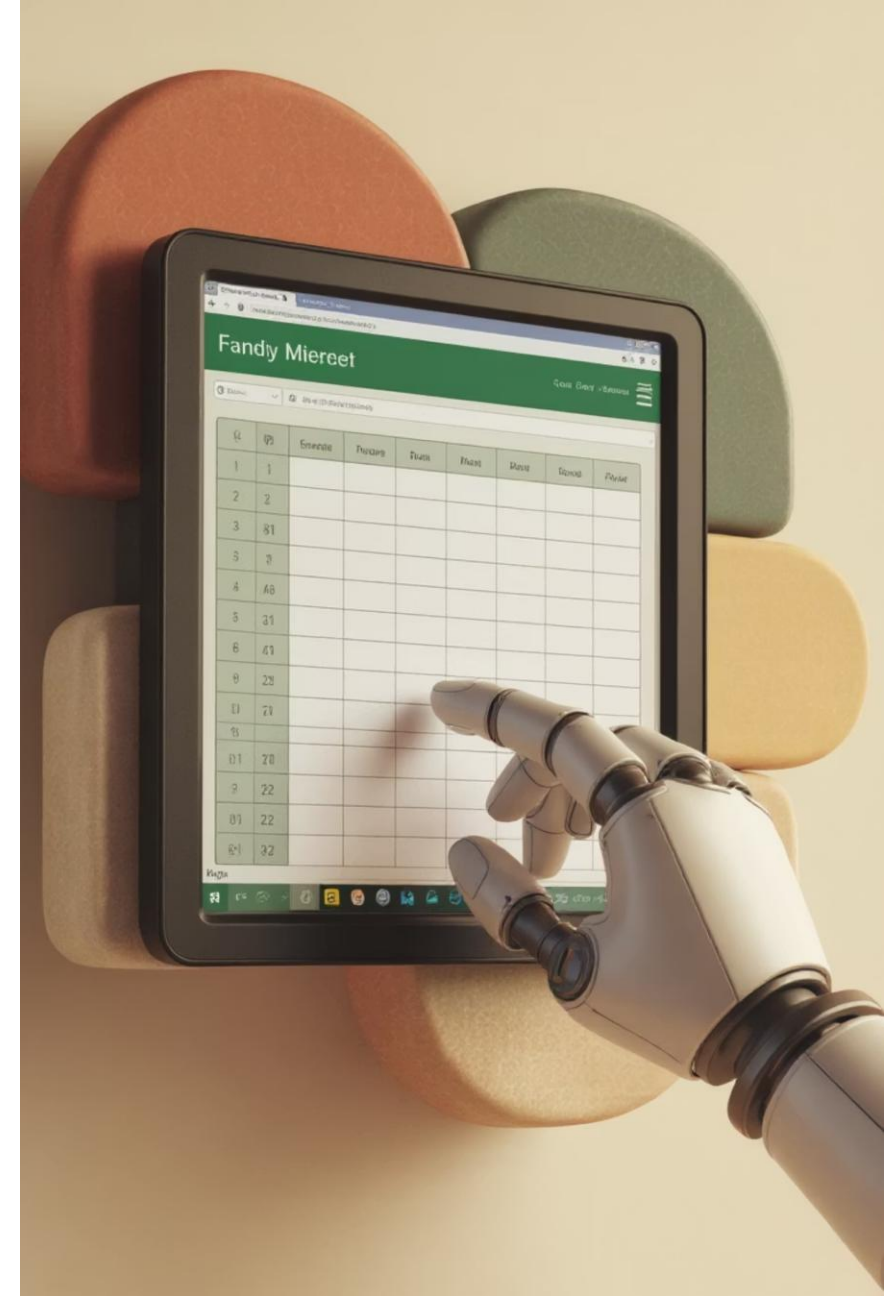
Claude Desktop (cloud Python) and **Claude Code** (local Python)

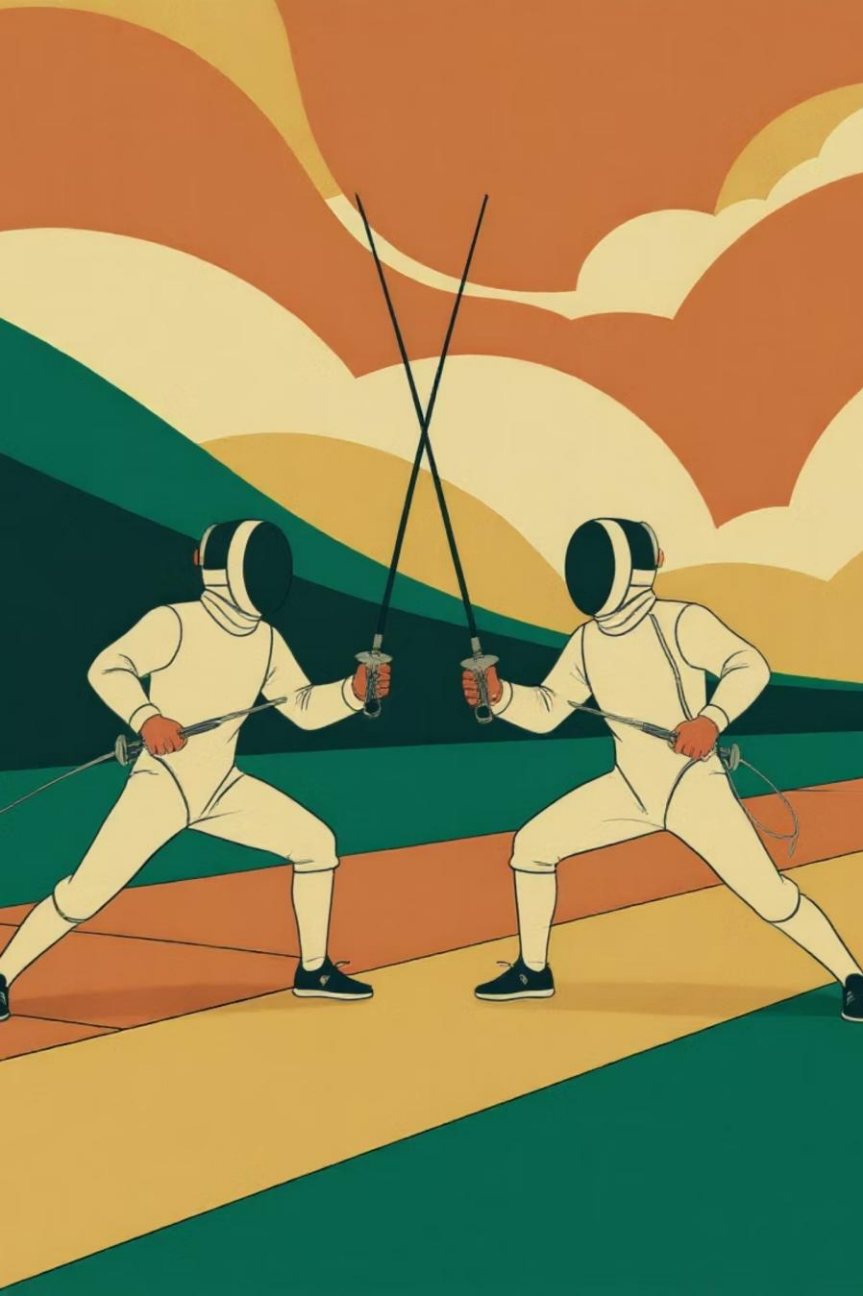
AI Can Create Excel

Claude became able to generate fully functioning, nicely formatted Excel workbooks in September 2025.

Microsoft has introduced the same Anthropic technology into Excel itself.

ChatGPT can also generate Excel files, but Claude seems to do a superior job at this time.





Claude Skills

- Recently introduced Claude skills are customized prompts saved as markdown files in a specific directory.
- Anthropic has created a number of skills. Users can easily create their own (ask Claude to do it!).
- Claude will consult the skill prompts as needed.
- Claude can generate Excel files because of Anthropic's new Excel skill. The markdown file (customized prompt) is here:

<https://github.com/anthropics/skills/blob/main/document-skills/xlsx/SKILL.md>

Claude Excel Example

HBS Case *Valuing Wal-Mart 2010*:



Read the uploaded case. Ignore the valuation method described in the case. Do a two-stage DCF valuation. Create a PowerPoint deck and fully functioning and professionally formatted Excel workbook.

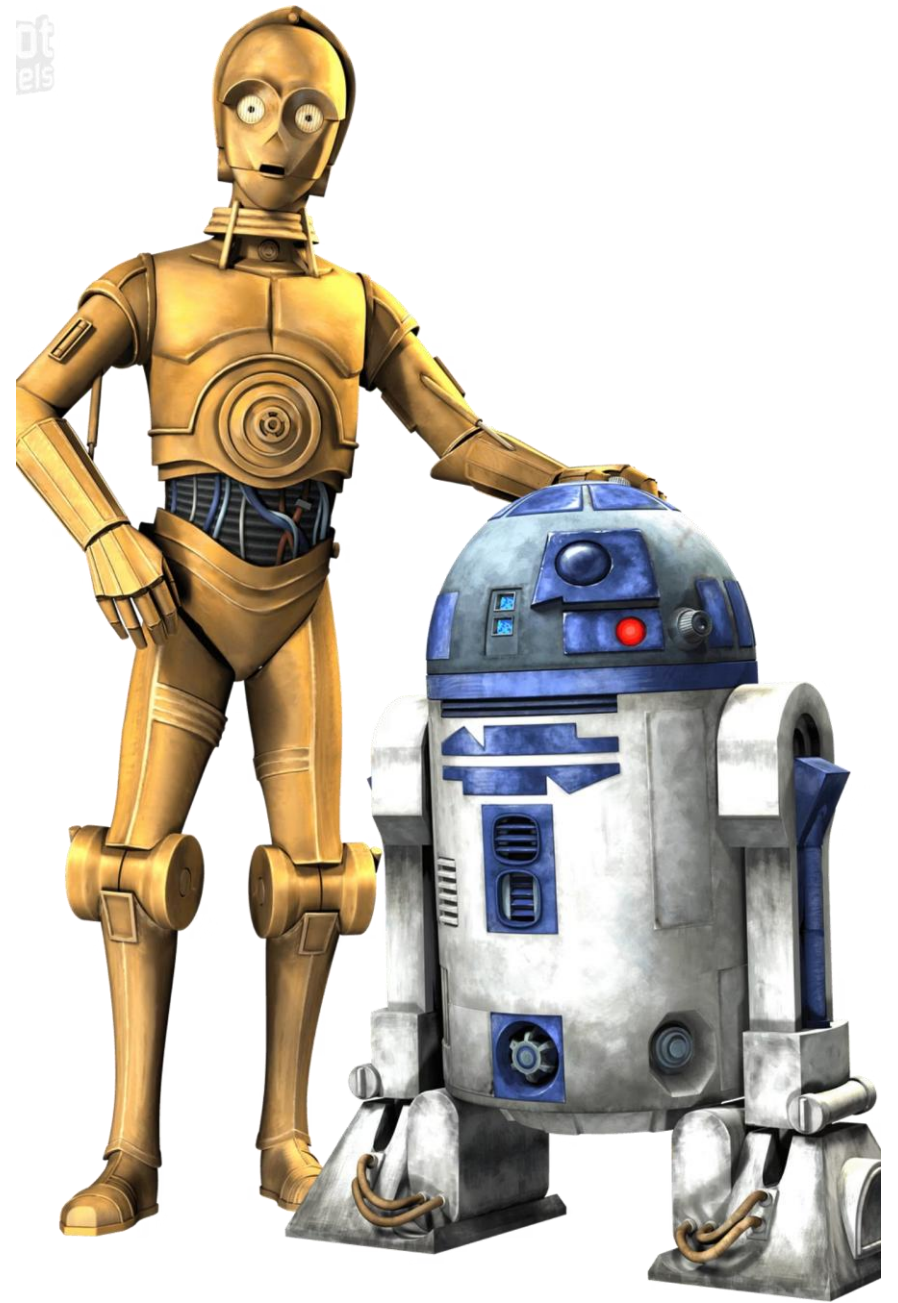
Document your assumptions and your reasons for them. Put years in columns and items in rows throughout the Excel workbook and PowerPoint deck.

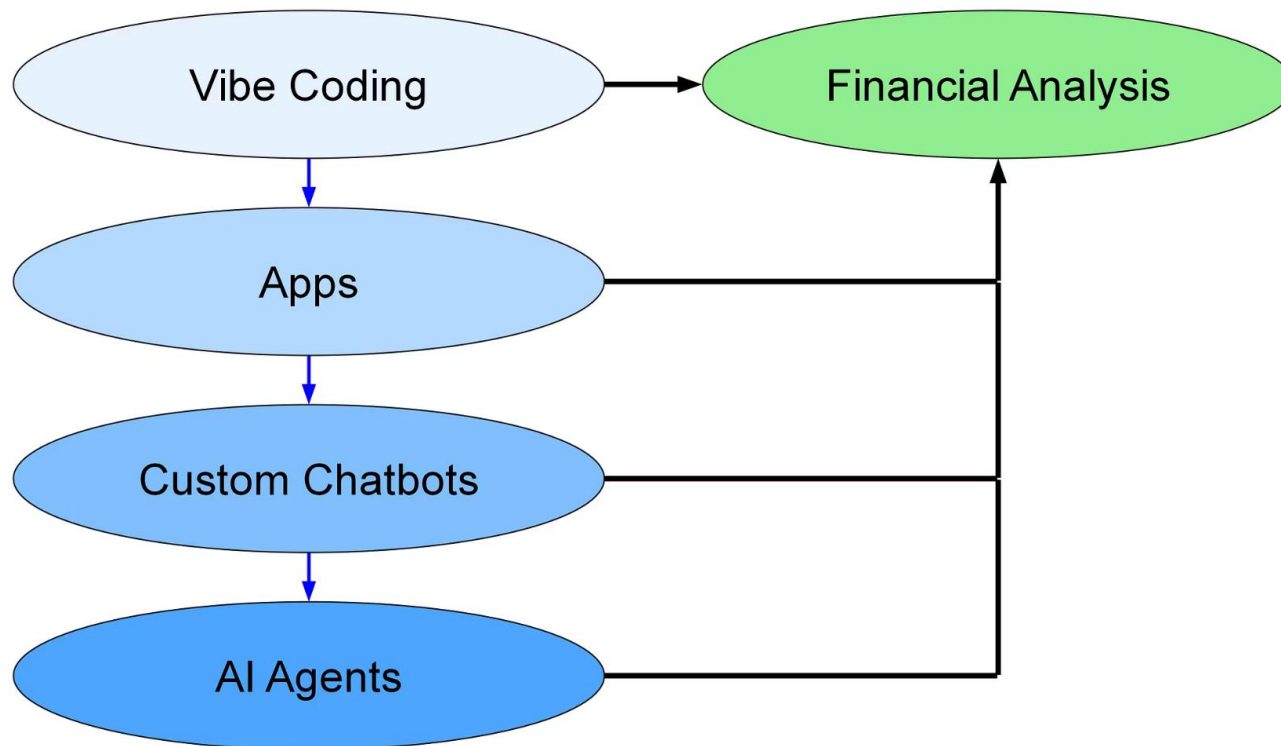
Microsoft's
Vibe Working

Powered by Claude

Agent Mode
in Excel

Chatbots and Agents





AI Plumbing

- Store code in an app
- Chatbots are apps
- Agents are chatbots with tools (tools=apps)

Why Teach the Plumbing?

- Students gain perspective on how things fit together in the AI space
- Students may end up at small or medium size firms and be the plumbers
- Building apps/chatbots/agents to do finance can deepen understanding of finance
- May check a box for recruiters

Why Teach the Plumbing?

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- Building apps/chatbots/agents to do finance can deepen understanding of finance
- May check a box for recruiters
- It's fun!**

Chatbot Elements

1 **User Interface**
The front-end interface where users interact with the system

2 **API Connection**
Secure connection to Large Language Model (OpenAI, Anthropic, Google, etc.)

3 **System Prompt**
Instructions that define the chatbot's behavior (on top of API system prompt)

4 **Document Retrieval**
Optional capability to access specific documents or knowledge bases

5 **Tool Integration**
Optional connections to external tools, databases, or APIs

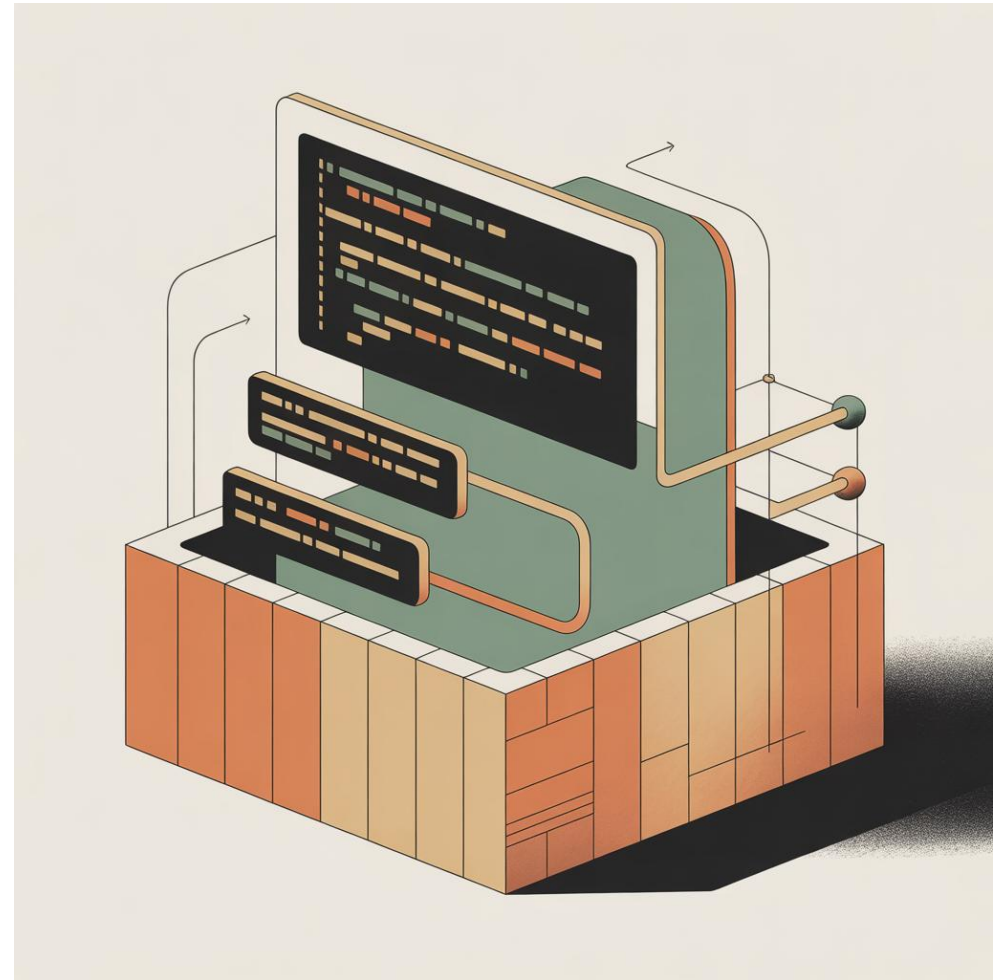
System Prompts: The Foundation

A system prompt is text sent to the LLM alongside each user prompt.

It defines the chatbot's personality, expertise level, response format, operational constraints, and provides supplemental information for the LLM.

The system prompt in a custom chatbot adds to the system prompt of the LLM's API.

Explore [Anthropic's System Prompts](#)



RAG: Retrieval Augmented Generation

Document Storage

Store relevant documents in a searchable database or vector store

Relevance Matching

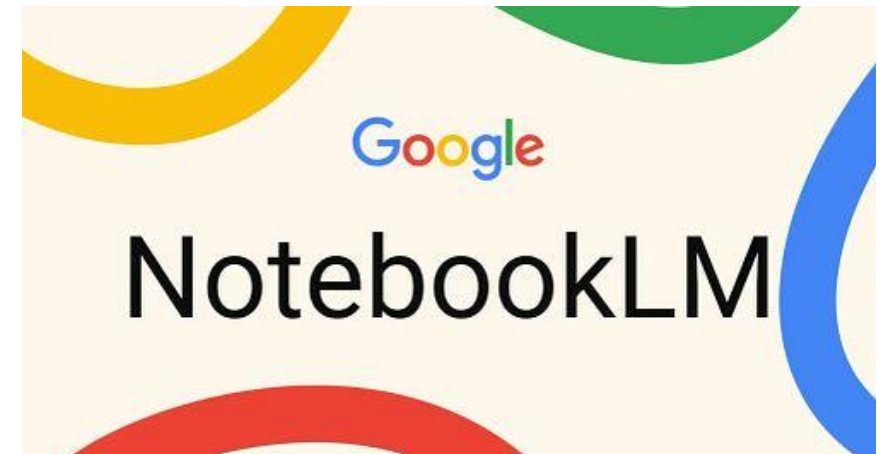
Choose most relevant document chunks using vector similarity algorithms

Context Integration

Send relevant chunks to LLM along with user prompt and system prompt

RAG is a key element of many corporate AI implementations.

An Easy-to-Use RAG Chatbot



Capabilities:

Chat

Audio overview

Video overview

Flash cards

Quiz

Reports

Mind map

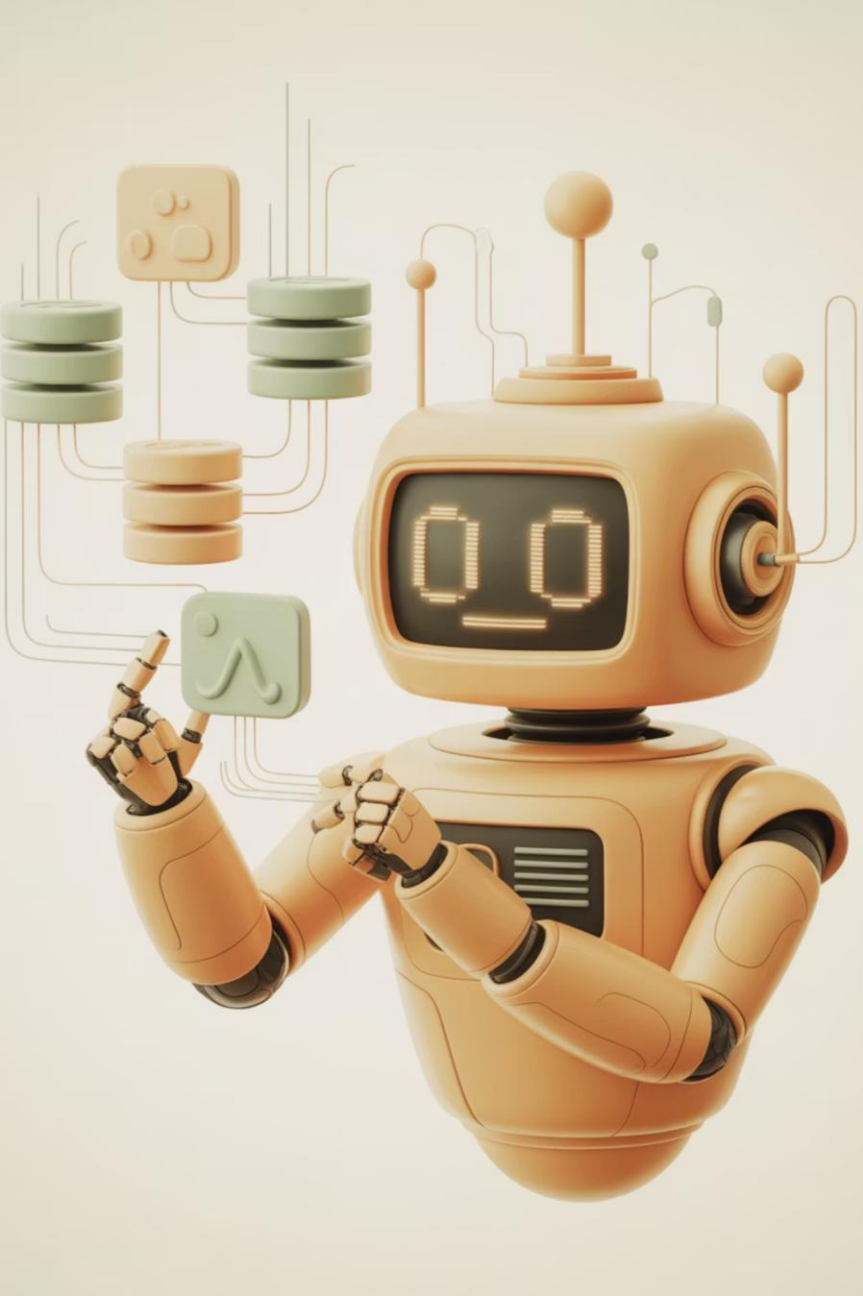
Easy sharing in university Google accounts, even available in Canvas.

More on NotebookLM later.

A Vibe-Coded RAG Chatbot

Visit chat.derivative-securities.org





Agents = Enhanced Chatbots

Definition

An AI agent is a chatbot equipped with tools to perform actions beyond conversation.

Database Integration

Database tools are particularly valuable for chatbots in financial applications.

Claude for Financial Services:

Anthropic: Through data providers, Claude (for Financial Services) has real-time access to comprehensive financial information.



Data Platforms

Box, Daloopa, Databricks



Financial Data

FactSet, Morningstar,
Palantir



Market Intelligence

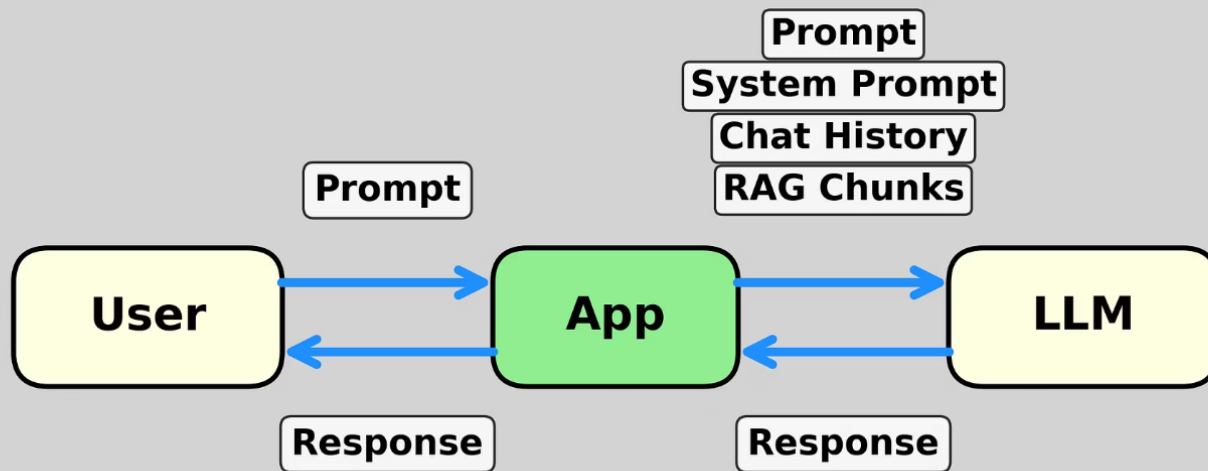
PitchBook, S&P
Global, Snowflake

Why MBAs Should Learn about Agents

Fortune, September 14, 2025:

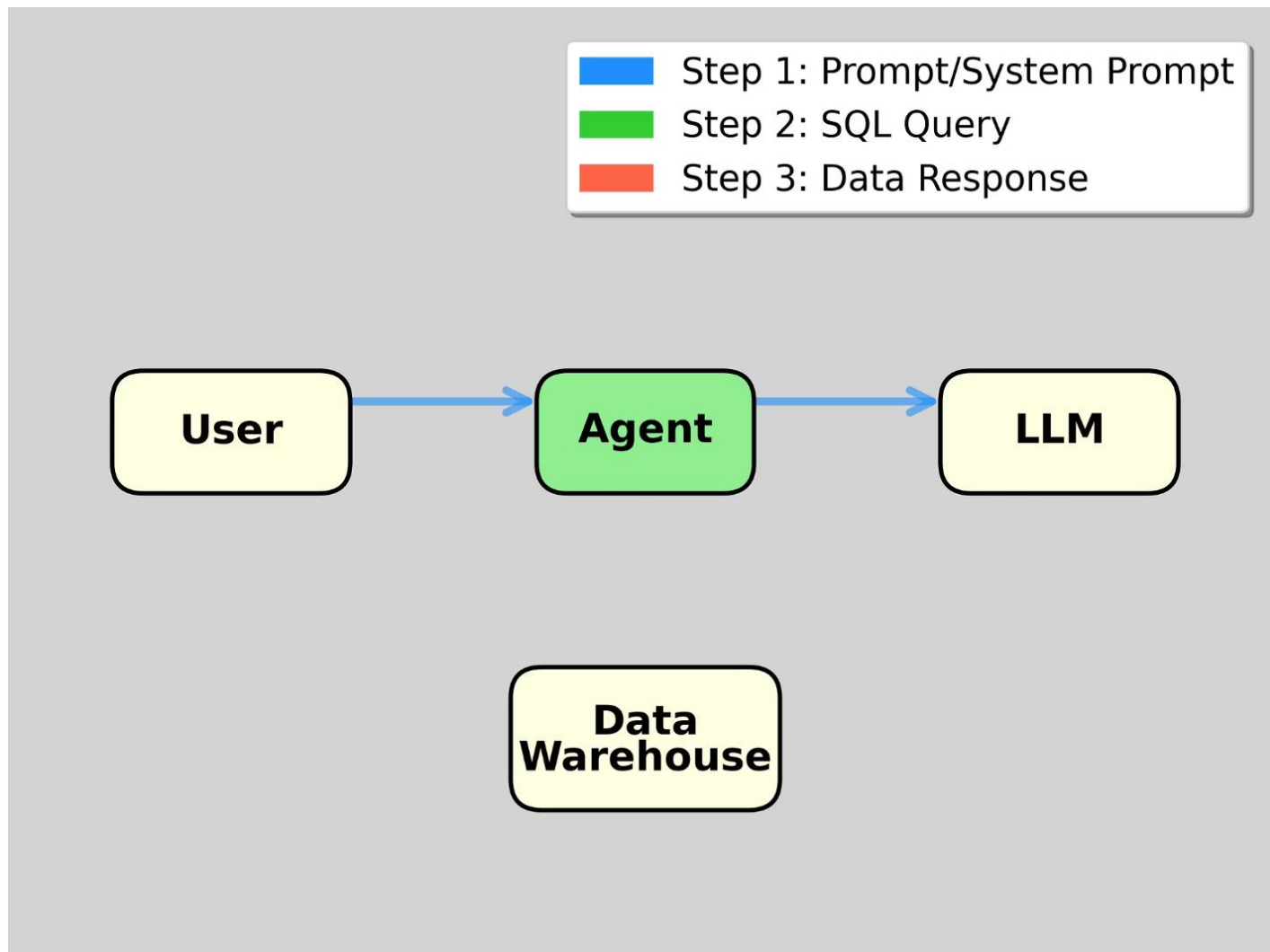
PromptQL, an enterprise AI platform created by San Francisco-based developer tooling company Hasura, is doling out **\$900-per-hour** wages to its engineers tasked with building and deploying AI agents to analyze internal company data using large language models (LLMs).

Tanmai Gopal, PromptQL's cofounder and CEO, said '**MBA types** ... are very strategic thinkers, and they're smart people, but they **don't have an an intuition for what AI can do.**'



Chatbot
Architecture

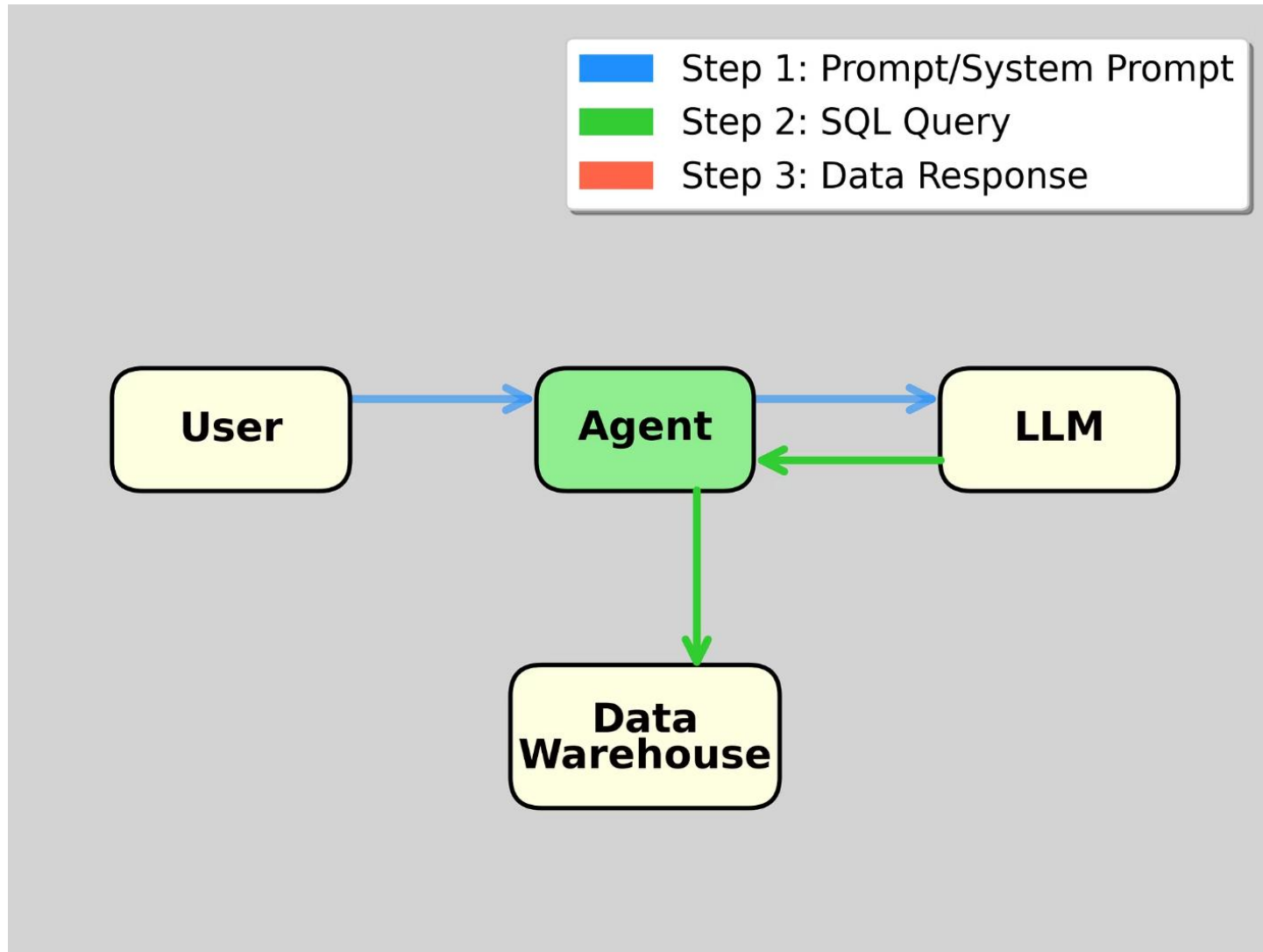
Database Agent



(1) User prompt → LLM

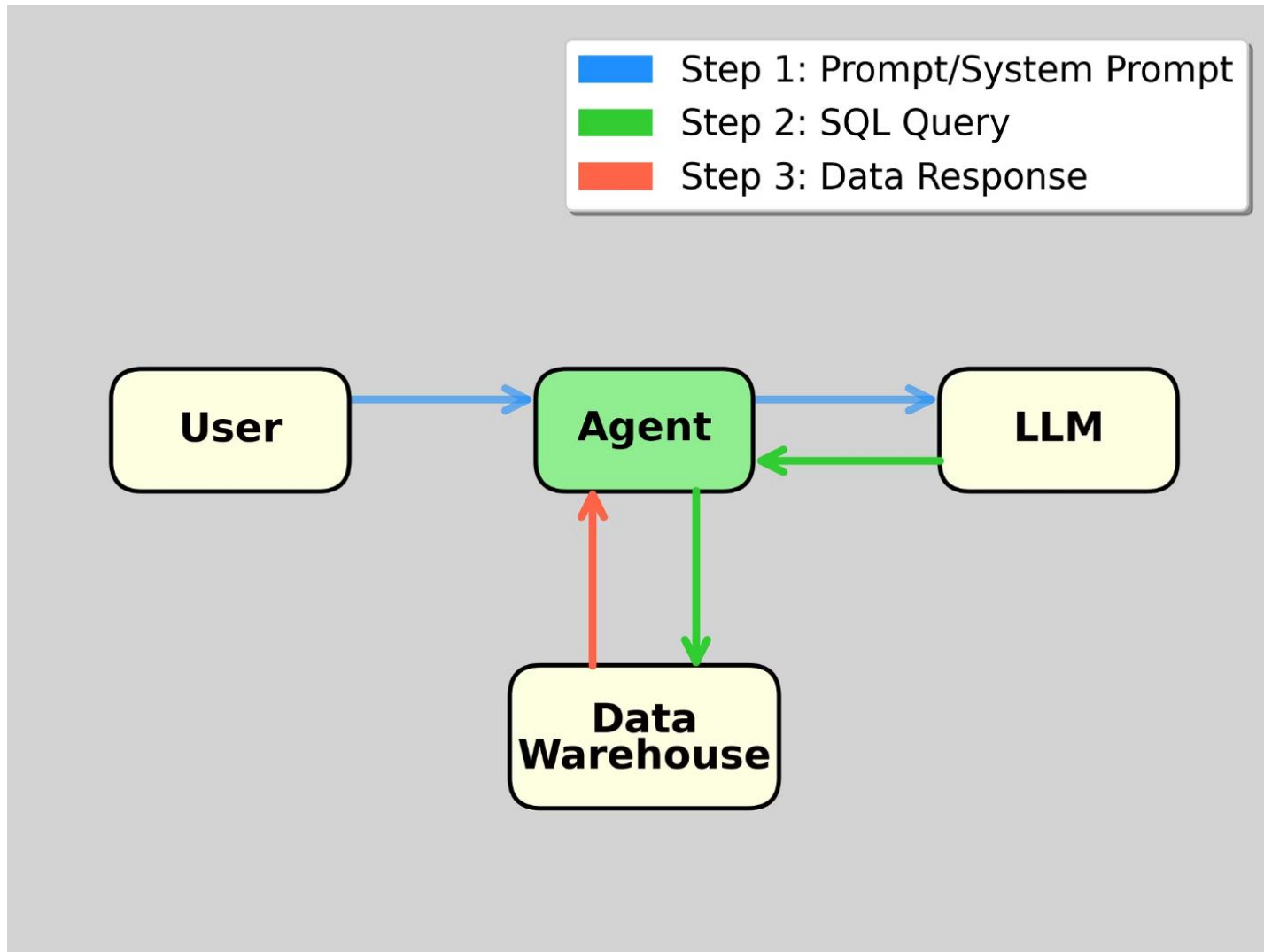
May be multiple rounds to clarify prompt

Database Agent



(2) SQL → Data Warehouse

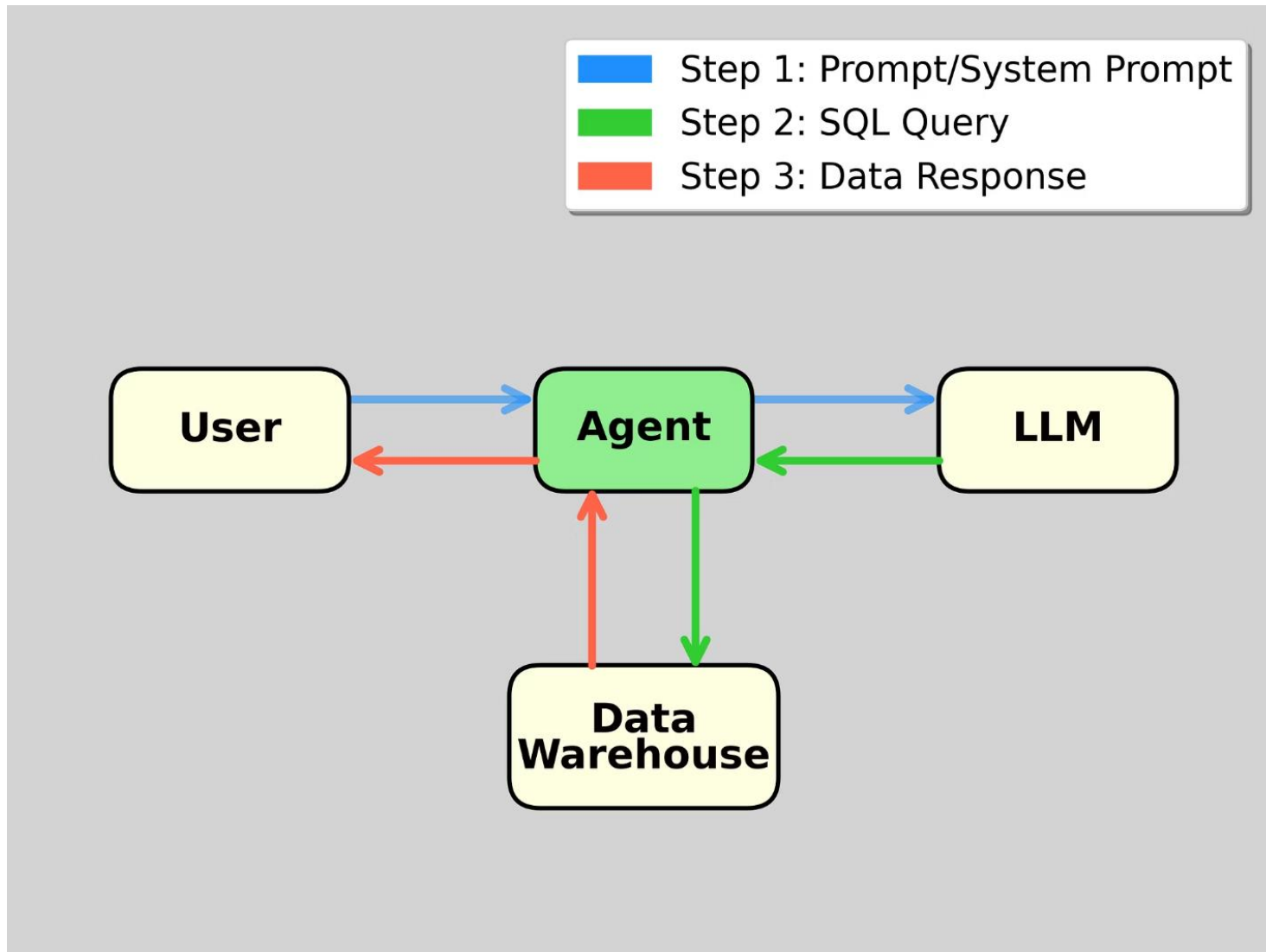
Database Agent



(3) Data or Error → Agent

Error message will be sent to LLM for new SQL

Database Agent



(3) Data → User

Could attach more tools.

For example, Python to analyze data.

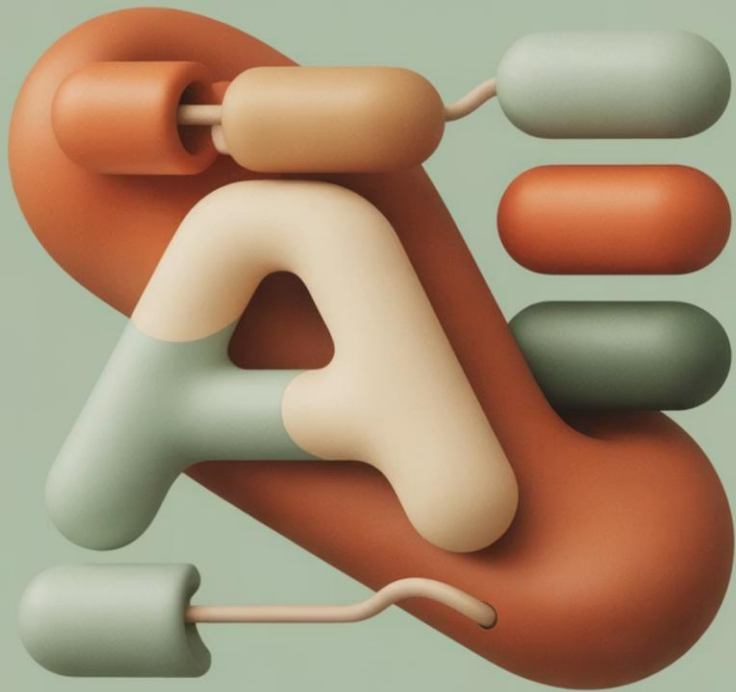
Then Output → User



Example: Rice Business Data Portal

[Visit data-portal.rice-business.org](https://data-portal.rice-business.org)

Demonstrates how students (and workers) can interact with complex databases using natural language.



MCP Servers

Anthropic's Model Context Protocol (MCP) enables adding custom tools to chatbots. I've added an MCP server for the Rice database to my Claude Code setup.

Demo Query: Get TSLA's ROE by quarter on a trailing 4-quarters basis for Q1 2020 through Q3 2025 **from the Rice stock database.**

Get TSLA's adjusted closing price starting Jan 1, 2020. Filter to keep only end-of-quarter prices.

Plot the ROE and filtered price series in the same figure, with ROE on the left y-axis and price on the right y-axis.



OpenAI



ChatGPT

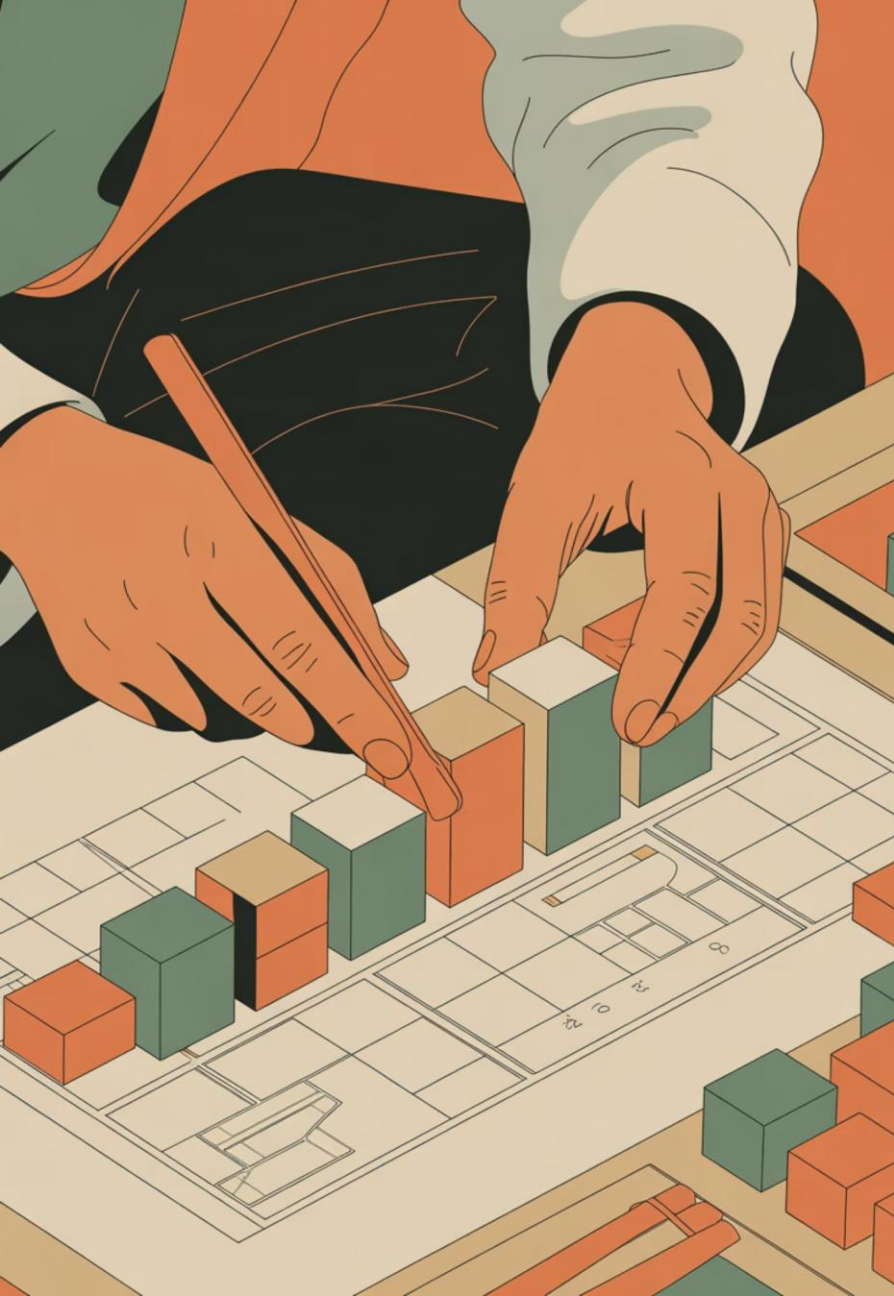
No-Code Solution: OpenAI Custom GPT

- System prompt
- RAG
- OpenAI tools: Web, Python, Dall-E
- External APIs and services



Deploys within ChatGPT (GPT Store)

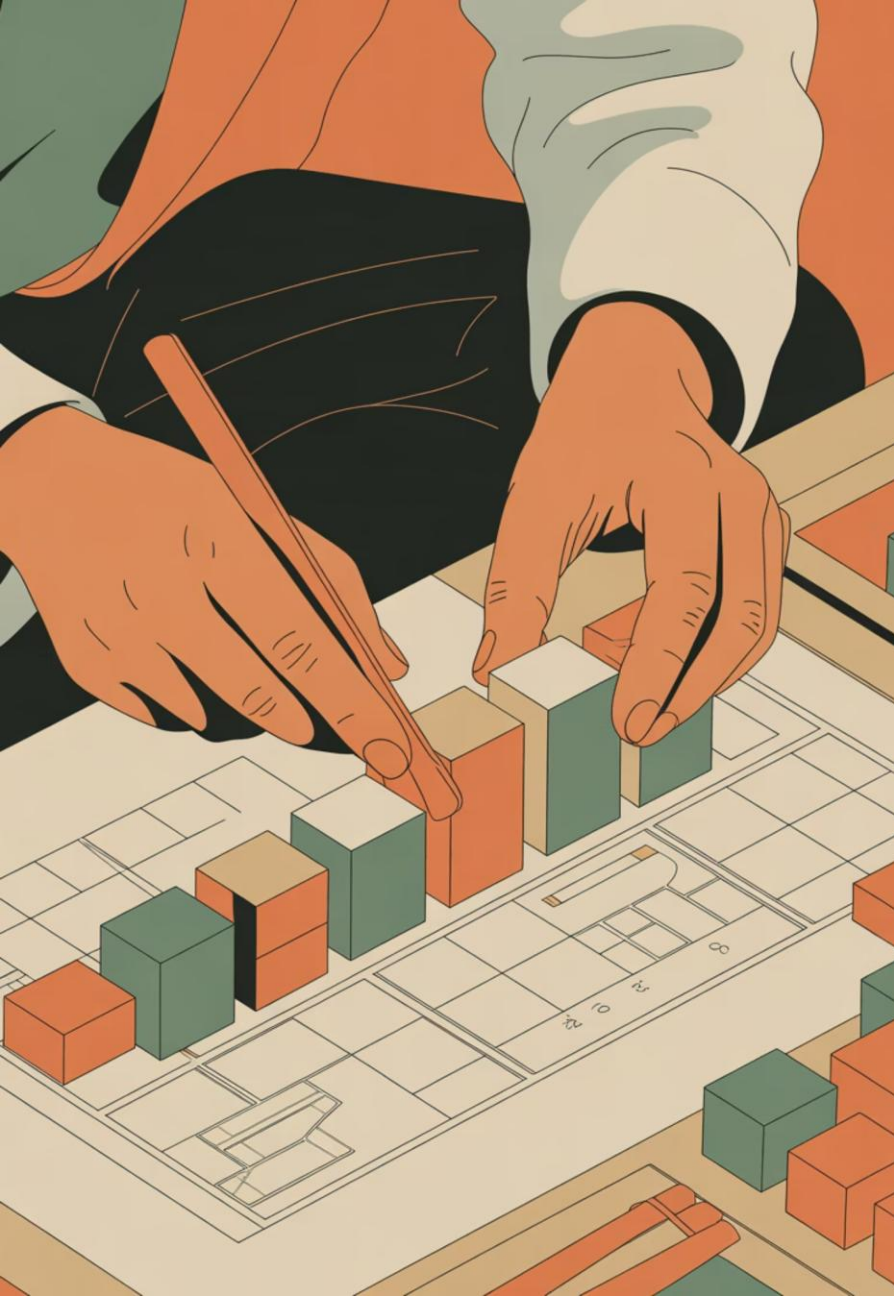
ChatGPT Plus subscription required for both creators and users



OpenAI's Agent Builder (AgentKit), released 10-6-2025

Visual canvas for building multi-step AI agent workflows without coding

Empowers users to visually assemble, debug, and export complex agent workflows



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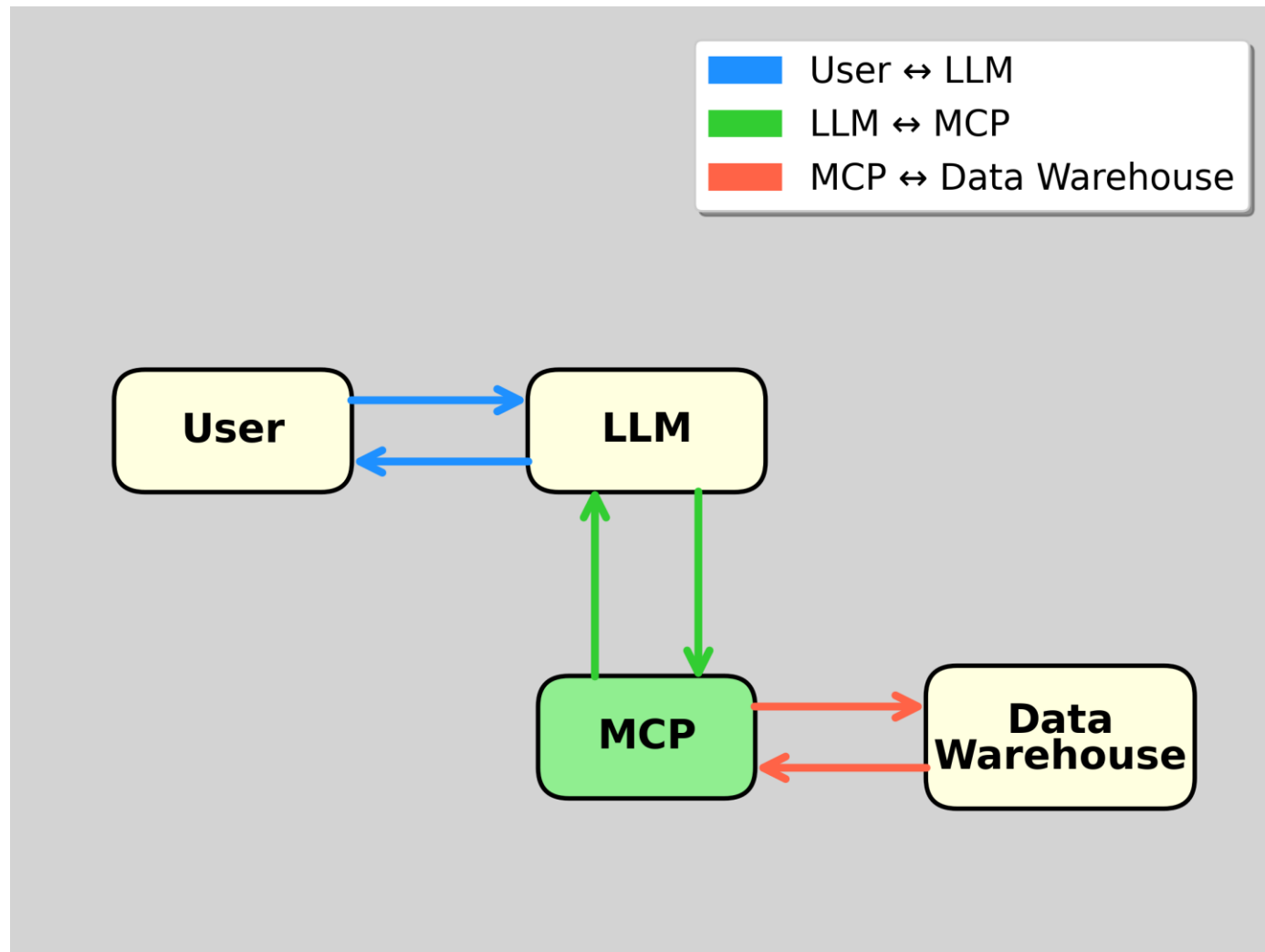
Empowers users to visually assemble, debug, and export complex agent workflows

Why point and click to build agents? Why not an agent to build agents?



BY ANTHROPIC

Model Context Protocol

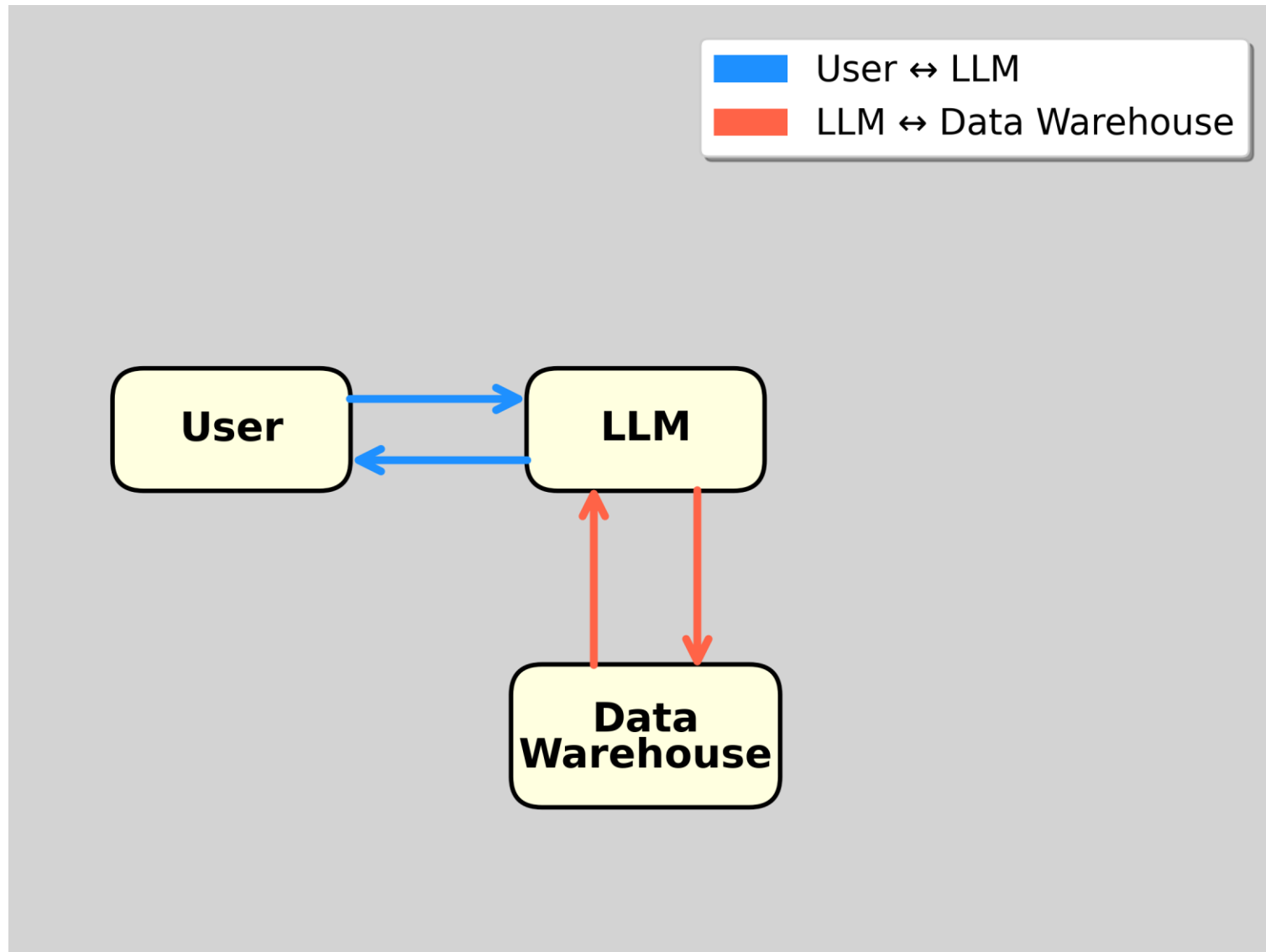


User chats directly with Claude

Agent logic embedded in MCP server (local or cloud)

User experience is Simplified. Use Claude for everything.

Claude Skills (10-16-2025)



User chats directly with Claude

Agent logic embedded in skill that Claude reads when invoked

Developer experience is Simplified

Claude has a skill at creating skills!

Claude Code with Skills

Simon Willison, 10-16-2025:

The skills mechanism is *entirely dependent* on the model having access to a filesystem, tools to navigate it and the ability to execute commands.

This is the biggest difference between skills and other previous attempts at expanding the abilities of LLMs, such as MCP and [ChatGPT Plugins](#).

Anything you can achieve by typing commands into a computer is something that can now be automated by Claude Code.





Previous Disruptive Technologies

Handheld Calculators

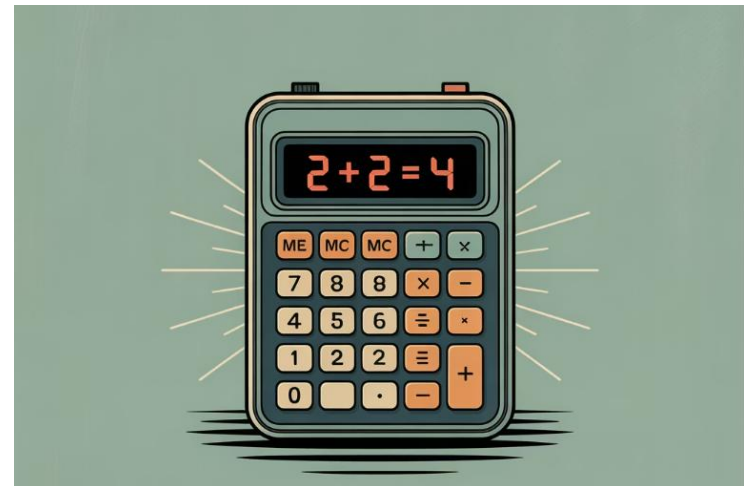
The Concerns

Educators worried students wouldn't learn fundamental arithmetic operations

The Reality

Studies haven't found significant negative effects on basic math skills. However, primary schools still provide practice and assessments without calculators.

Some hoped schools could move beyond calculation to teach deeper mathematical concepts. That didn't happen either.





Personal Computers & Spreadsheets

We added courses on how to use spreadsheets

We updated existing courses to use spreadsheets

We assess how well students can use spreadsheets (no pencil and paper valuation analyses)

Teaching how to do finance in spreadsheets is a main educational goal.

Why Spreadsheets Succeeded

-  **Business Relevance**
Spreadsheets are extensively used in real business environments
-  **Educational Value**
Ideal for teaching the logic of valuation and capital budgeting concepts
-  **Tool Evolution**
Spreadsheets don't automatically do what we want students to learn

AI: A Different Challenge

✓ Business Usage

AI is increasingly used across business functions

✓ Educational Tool

AI provides excellent tutoring and learning support

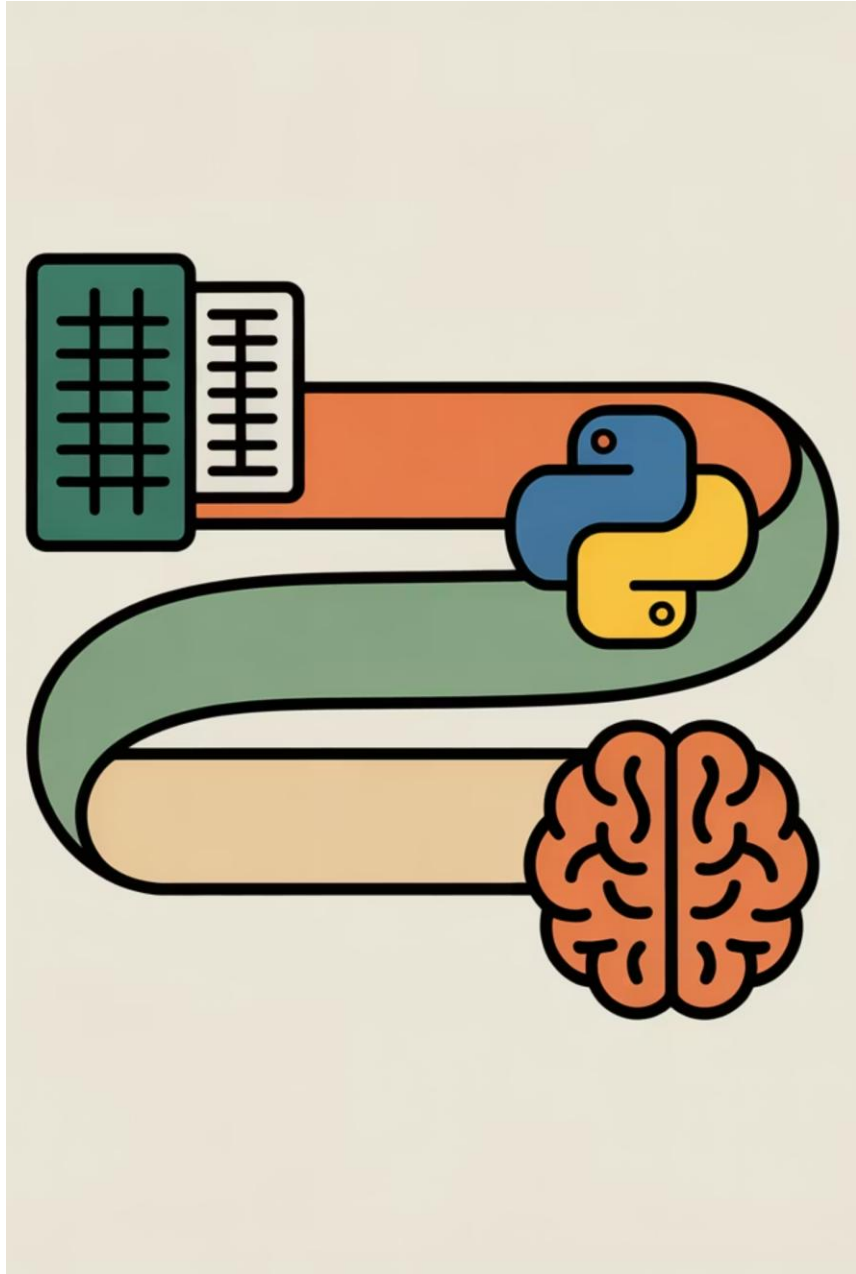
✗ Automation Risk

AI can automatically perform many tasks we previously taught manually

AI is more like calculators than spreadsheets. We face similar concerns about skill acquisition. Like primary schools with arithmetic, we'll need practice and assessments without AI to ensure fundamental competencies.

AI in Finance Courses





The New Skills Landscape

Brookings Institute, 2025:

As AI models begin to handle underwriting, compliance, and asset allocation, the traditional architecture of financial work is undergoing a fundamental shift.

As job descriptions evolve, so does the definition of financial talent. **Excel is no longer a differentiator. Python is fast becoming the new Excel.**

But technical skills alone will not cut it. The **most in demand profiles today are those that speak both AI and finance.**

A New Teaching Sequence

1

Concepts First

Begin with fundamental concepts using slides or whiteboard as usual

2

Tool Implementation (Maybe?)

Demonstrate implementation in traditional tools (usually Excel) as usual

3

AI Enhancement

New step: **Apply concepts using chatbots and AI agents**

Traditional Approach

Teach Excel implementation → students build Excel models

Additional Piece

Teach AI implementation → **students conduct AI chats or build AI apps**

Straight from Slides/Whiteboard to AI?

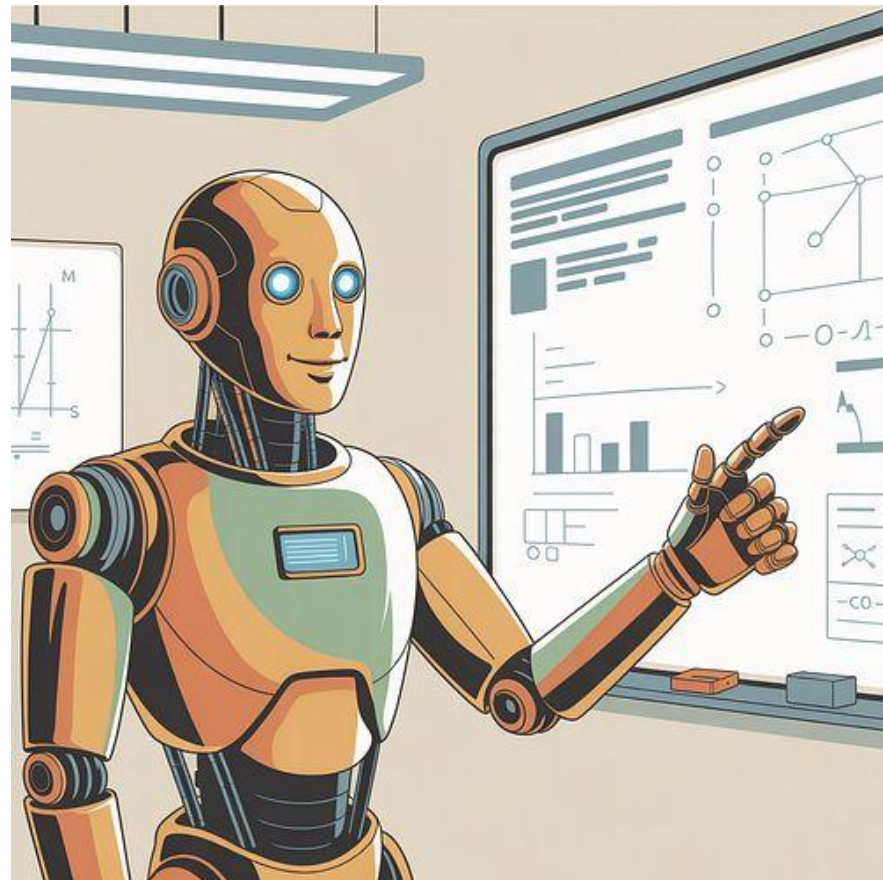
Linear regression

Simulation

Mean-variance Optimization

Option valuation

DCF analysis



Straight from Slides/Whiteboard to AI?



Linear regression



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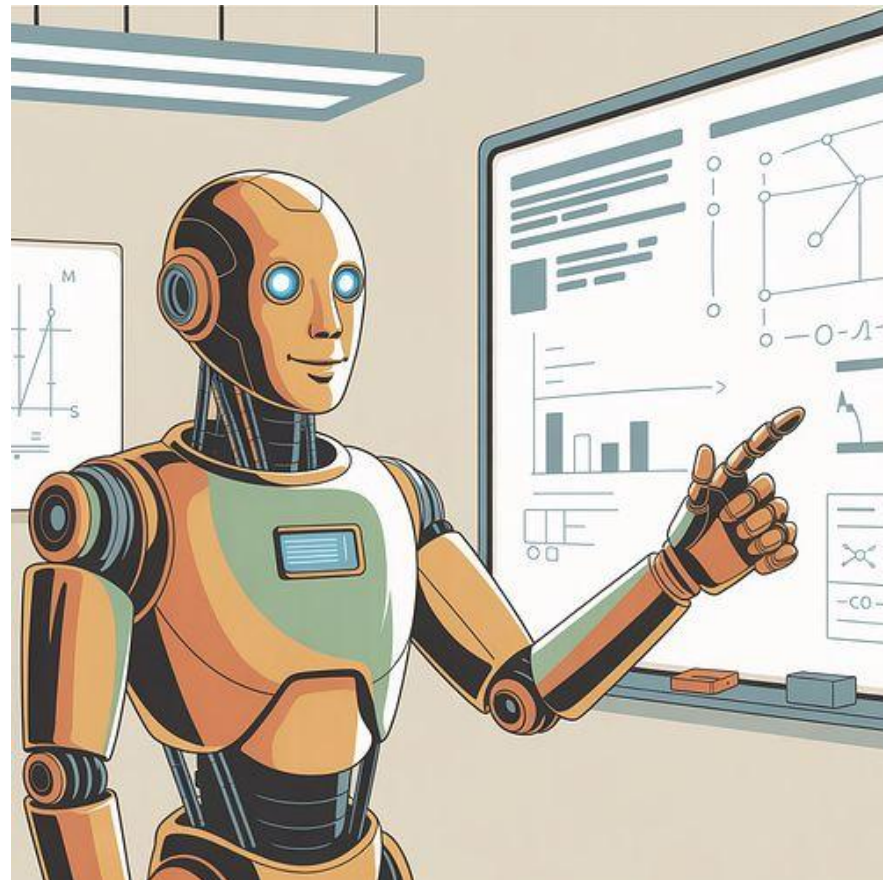
Mean-variance Optimization



Option valuation



DCF analysis





AI in Education in General

Three Legs of Assessment

Assignments

"Turn in your chat, its output, and your evaluation of the output"

Exams Without AI

In-class assessments ensuring students understand fundamental concepts independently

Exams With AI

More ambitious assessments leveraging AI assistance - complex cases can become exams with AI support

AI as Educational Partner

AI can be a highly effective tutor, and we should leverage it in our teaching.

Replace Reading Assignments

Ask students to chat with chatbots instead of traditional reading assignments



Pre-Class or In-Class

Assign pre-class or in class before slides and class discussion.

Assignment: "Ask your chatbot to teach you about [topic]. Tell it to ask you questions to ensure you are understanding."

If Chatbot Reliability is a Concern

Use RAG

Build custom systems (could use Google's Notebook LM or Custom GPT) for specific source control

Build with AI Reports

Can have AI create a comprehensive report, then edit if needed and use for RAG or in system prompt

Reliability Reality

LLMs today are significantly more reliable than earlier versions - hallucinations have diminished

NotebookLM Example

Step 1:

I asked ChatGPT to prepare a detailed report on binomial option in markdown format. I could have used a different chatbot.

Link to ChatGPT conversation:

<https://chatgpt.com/share/68e7cbf4-013c-8006-89bb-1166720463ff>

Step 2:

I uploaded the markdown file to NotebookLM and asked it to generate flash cards and a video overview.

Link to NotebookLM notebook:

<https://notebooklm.google.com/notebook/7e3e40f0-a937-46cb-8eea-7c713bdcb8d4>

Link to NotebookLM video:

https://genai4finance.kerryback.com/Binomial_Option_Model.mp4

Effect of AI on Junior Employment

Harvard economists' study:

AI is reducing
• employment of junior workers but not senior workers

[Lichtinger and Maasoum](#) (20,000+ SSRN downloads in one month)

JPMorgan quote:

"Many, many of the
• processes underneath are being done by AI."

Junior professors?

[Scott Latham, UMass, Chronicle of Higher Education:](#)

[Are you Ready for the AI University?](#)



Deep Dive into Claude Code

Past, Present, & Future of UI's

- **Terminal Era:** Must know Unix, MS-DOS, Linux, ...
- **Graphical User Interface:** Point & click
- **Natural Language Chat:** AI agent does everything



Claude Code

Part of a new generation of terminal-based coding agents, along with OpenAI Codex and Gemini CLI.

It integrates well with popular development environments like VS Code.

Built on standard Anthropic models, Claude Code appears to have an exceptionally well-designed system prompt that optimizes it for development tasks.



Requires Claude Pro subscription

Claude Code's Capabilities



File Operations

Create, edit, copy, move, and delete files



Document Creation

Generate and edit **LaTeX** documents.
Generate **Word** docs and **PowerPoint** presentations



Python Development

Write, edit, and execute **Python scripts** and **Jupyter notebooks**



Deployment

Create **GitHub** repositories and **deploy apps** on hosting platforms like Hugging Face Spaces, Streamlit Community Cloud, Koyeb, and Render

OpenDoor CEO: Default to AI

So, starting today the first line in everyone's job expectation is simply this: Default to AI. (This applies to everyone, including me!)

Starting with the next performance review, in addition to asking how much impact each employee delivered, we will also ask ourselves how frequently does each person default to AI.

If you reach for a Google Doc or Sheets before you reach for an AI tool, you are not defaulting to AI. If the prototype is not built in Cursor or Claude Code, you are not defaulting to AI.

— *OpenDoor CEO, Oct 2025*

Claude Code Demo Part 1

Create a Python script for mean-variance optimization.

When run, it should prompt the user for the number of assets, their means, standard deviations, and correlations and the risk-free rate.

Only ask for the minimum number of correlations required to compute the correlation matrix.

Assume short sales are allowed.

Compute and display the tangency portfolio.



Claude Code Demo Part 2

Create a Streamlit app from the mean-variance code.

I want the app to create a Word doc containing the means, standard deviations, and correlations of the assets and a description of the tangency portfolio.

Tunnel the app using my NGROK_TOKEN stored at C:\Users\kerry\Dropbox\ENV



Streamlit

Claude Code Demo Part 3

Modify the Streamlit app as follows:

Use my OpenAI_API_KEY stored at
C:\Users\kerry\Dropbox\ENV.

Send the Word doc to GPT 5 and ask for an
analysis of why the tangency portfolio is what
it is.

Revise the Word doc to include the GPT
analysis.

Tunnel the app using my NGROK_TOKEN
stored at C:\Users\kerry\Dropbox\ENV



Claude Code Skill Example

Claude Prompt 1: Download and install the Claude skill for creating Word docs

Claude Prompt 2: Download and install the Claude skill for creating skills

Claude Prompt 3: Create a mean-variance-optimization skill.

It should prompt the user for the number of assets and their expected returns, standard deviations and correlations (the minimum number of correlations needed to compute the correlation matrix), and for the risk-free rate.

Then calculate the tangency portfolio, allowing short sales.

Then create a Word doc containing the relevant data, the tangency portfolio, and your explanation of why the tangency portfolio is what it is.

Claude Code Skill Example

Claude Prompt 1: Download and install the Claude skill for creating Word docs

Claude Prompt 2: Download and install the Claude skill for creating skills

Invoke and use the skill (probably /mean-variance-optimization).

Claude Prompt 4: Review the conversation and improve the mean-variance-optimization skill.

Claude Prompt 3: Create a mean-variance-optimization skill.

It should prompt the user for the number of assets and their expected returns, standard deviations and correlations (the minimum number of correlations needed to compute the correlation matrix), and for the risk-free rate.

Then calculate the tangency portfolio, allowing short sales.

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When Should Students Learn Claude Code

(Or Whatever the Current Best AI Tool is)

- First term as a tools course (or in Data Analysis, for example)
- Last term as an advanced elective



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- Last term as an advanced elective



When Should Students Learn Claude Code

(Or Whatever the Current Best AI Tool is)

- First term as a tools course (or in Data Analysis, for example)
- Last term as an advanced elective

HBS licensed Julius.ai to use in Data Analysis course.



Domain Knowledge will be
More Important than Ever

**AI will level the playing field for tool
skills (assuming students can use AI).**

**That will make domain knowledge
more important than ever as a
differentiator.**





Some Additional Resources

Scan the QR code to access
genai4finance.kerryback.com